

Problem 7.1 (MATLAB) **Multivariate Multiple Linear Regression (MMLR)**

The file '***dataset_problem_7_1.mat***' contains measurements of properties of pulp fibers and the paper made from them. There are a total of 62 observations.

The multivariate (vector) response variable is $Y = [Y_1 \ Y_2 \ Y_3 \ Y_4]$, where Y_1 = paper breaking length, Y_2 = elastic modulus of the paper, Y_3 = paper stress at failure and Y_4 = burst strength of paper.

The four predictor variables is z_1 = arithmetic fiber length, z_2 = long fiber fraction, z_3 = fine fiber fraction and z_4 = zero span tensile.

It is desired to build and examine a multivariate multiple regression model on the form

$$[Y_1 \ Y_2 \ \dots \ Y_m] = [1 \ z_1 \ \dots \ z_r] \begin{bmatrix} \beta_{01} & \beta_{02} & \dots & \beta_{0m} \\ \beta_{11} & \beta_{12} & & \beta_{1m} \\ \vdots & \vdots & & \vdots \\ \beta_{r1} & \beta_{r2} & \dots & \beta_{rm} \end{bmatrix} + [\varepsilon_1 \ \varepsilon_2 \ \dots \ \varepsilon_m]$$

where, as explained, $m = 4$, $r + 1 = 5$ and $n = 62$

- Load the data file
- Calculate an unbiased Least Squares estimate of the regression coefficient matrix
- Calculate an unbiased estimate of the (m x m) covariance matrix $\Sigma_\varepsilon = \text{Cov}[\varepsilon]$
- Perform an uncorrected (not mean-corrected) decomposition of (m x m) SS matrices: $SST = SSR + SSE$
- Perform an mean-corrected decomposition of (m x m) SS matrices: $SST = SSR + SSE$
- Calculate the "coefficient of determination", R_{sqr} , and the "adjusted coefficient of determination", $R_{\text{adj_sqr}}$, using the usual univariate definitions of these but substituting the Sums of Squares in the equations with determinants of the multivariate (m x m) matrix versions of the SS's
- Next, for inference for the regression coefficients, indicate the ((r+1) x 1) vector of estimated regression coefficients for Y_1
- Estimate the ((r+1) x (r+1)) covariance matrix for these coefficients
- Calculate 95% simultaneous confidence intervals for regression coefficients for Y_1
- Calculate 95% Bonferroni confidence intervals for regression coefficients for Y_1

- Perform an Extra Sum of Squares based test for significant regression model (significant dependency of $Y = [Y_1 \dots Y_m]$ on $[z_1 z_2 \dots z_r]$), that is test the hypothesis H_0 : all regression coefficients = 0 except the m intercept coefficients
- Perform an Extra Sum of Squares based test for significant predictor variable z_1 (significant dependency of $Y = [Y_1 \dots Y_m]$ on z_1), that is test the hypothesis H_0 : all regression coefficients for $z_1 = 0$
- Perform similar tests for significant predictor variables z_2, z_3 and z_4
- Calculate 95% simultaneous prediction intervals for $E[Y(z_0)]$, the mean of new vector responses at $z=z_0$, for $z_0 = [z_1 z_2 z_3 z_4] = [0.33 \ 45.5 \ 20.375 \ 1.01]$
- Calculate 95% simultaneous prediction intervals for $Y(z_0)$, a single new vector response at $z=z_0$
- Perform univariate model check: analysis of the residuals (estimated model errors) corresponding to each of the response variables. Plot the residuals as function of observation index and response variable. Also plot a histogram and a QQ-plot for each of the residuals
- Perform multivariate (simultaneous) model check as to whether the residual vectors are likely to stem from a multivariate normal distribution. Make matrices of scatter plots and investigate the Mahalanobis distances
- Conclude on the validity of the model
- Finally, use the built-in function **mvregress** to create the MMLR model (you will have to define a cell array, see documentation), estimate the regression coefficients matrix and estimate the $(m \times m)$ covariance matrix $\Sigma_\varepsilon = \text{Cov}[\varepsilon]$